

WWI Sales Analysis: Data Quality Issues Summary

Fact Sale Dataset — Wide World Importers | 26,397 rows, 21 columns

#	Issue	Finding	Decision
1	Date format	Clean, except for 13 missing delivery dates	Converted to date format
2	Missing delivery date	13 rows (all on the last invoice date)	Rows kept, flag added
3	Customer Key = 0	9,077 rows (34.4%)	Unchanged, documented
4	Tax Amount difference	253 rows have a fixed -0.01 difference	Unchanged, caused by rounding
5	Negative Profit	566 rows (2.14%), 87% from one product	Flag added, reported to business
6	IQR outlier test	6–12% false outlier warnings	Not used
7	Missing dimension tables	Salesperson and City show IDs only	Analysis done using IDs

Detailed Explanations

1. Date Format

- Finding:**

The `Invoice Date Key` and `Delivery Date Key` columns were stored as text (e.g., "10/22/2013") instead of proper dates. A check was done to see if any delivery date happened before the invoice date, but 0 errors were found.

- Decision & Rationale:**

Both columns were converted to date format using pandas. This was necessary to filter by dates, group by months, and use time functions in Power BI.

2. Missing Delivery Date

- **Finding:**

The `Delivery Date Key` column had 13 missing values. Looking closely, all 13 rows belonged to the last invoice date in the dataset (2016-05-31).

- **Decision & Rationale:**

These rows were not deleted. Instead, an `Is Delivered` column was added. The simplest explanation is that these orders were not delivered yet when the data was collected—this is a normal business situation, not a data mistake.

3. Customer Key = 0

- **Finding:**

9,077 rows (34.4% of all data) have `Customer Key` and `Bill To Customer Key` set to 0. This happened mostly with specific salespeople, spread evenly over all years (2013–2016), and the average order value was almost identical to regular sales (745.70 vs 756.98).

- **Decision & Rationale:**

The values were left as 0 because changing an ID number to an arbitrary value would be misleading. This was noted as "over-the-counter or retail sales without a registered account"—a business process, not a system error.

4. Tax Amount Difference

- **Finding:**

In 253 rows (~1%), the recorded `Tax Amount` was exactly 1 cent (0.01) lower than expected from the standard tax formula ($\text{Total Excluding Tax} \times \text{Tax Rate}$). All mismatches happened at the 15% tax rate, and the difference was identical in every row.

- **Decision & Rationale:**

The original numbers were kept. The consistent pattern shows that the main system simply uses a slightly different rounding rule. Because 1 cent per row has almost zero financial impact, no changes were made.

5. Negative Profit

- **Finding:**

566 rows (2.14%) lost money. 87% of these loss-making rows (492 rows) came from just one product group: *Halloween Zombie Mask (Light Brown)* across 4 sizes. **Every single sale** of this product lost money (selling price was 18, but cost was 19).

- **Decision & Rationale:**

The sales figures were kept as they are to show true company data. A `Sold Below Cost` flag column was created, and the issue was reported to the team as a pricing mistake.

6. IQR Outlier Test

- **Finding:**

A standard statistical test (IQR) was run on Quantity, Price, Total, and Profit. It flagged 6% to 12% of the dataset as "outliers" (thousands of rows).

- **Decision & Rationale:**

The test results were ignored and no data was removed. Sales data naturally has large bulk orders, so the test falsely marked valid sales as errors. Logical business checks were used instead.

7. Missing Dimension Tables

- **Finding:**

The `Salesperson Key` and `City Key` columns contain only ID numbers; there are no separate tables with actual employee names or city names.

- **Decision & Rationale:**

The data was kept as is, and analysis was done using IDs (e.g., "Salesperson 19"). Adding real name tables from the CRM/HR system is recommended for future reports.